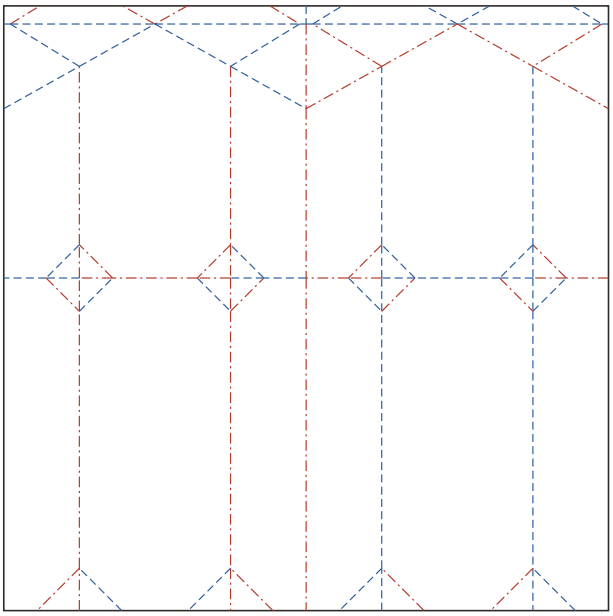
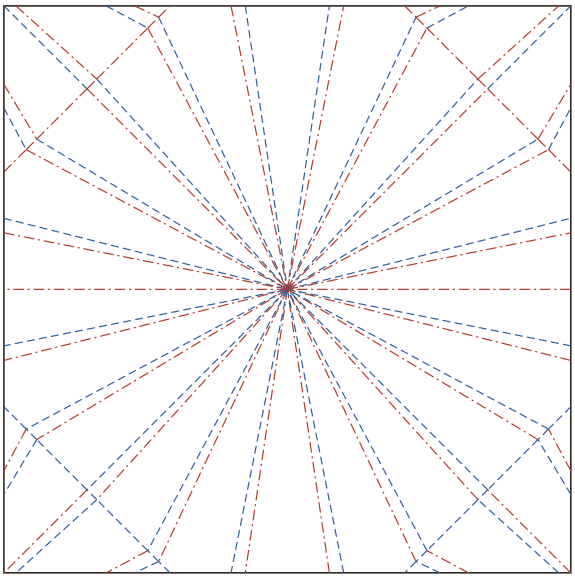
Paper Couture · Zori and folding fan · crease patterns

Actual size: print at 100% (no scaling). The next two pages give each at twice the size, for a first try; it folds the same.

Zori · 8 cm



Folding fan · 7.5 cm



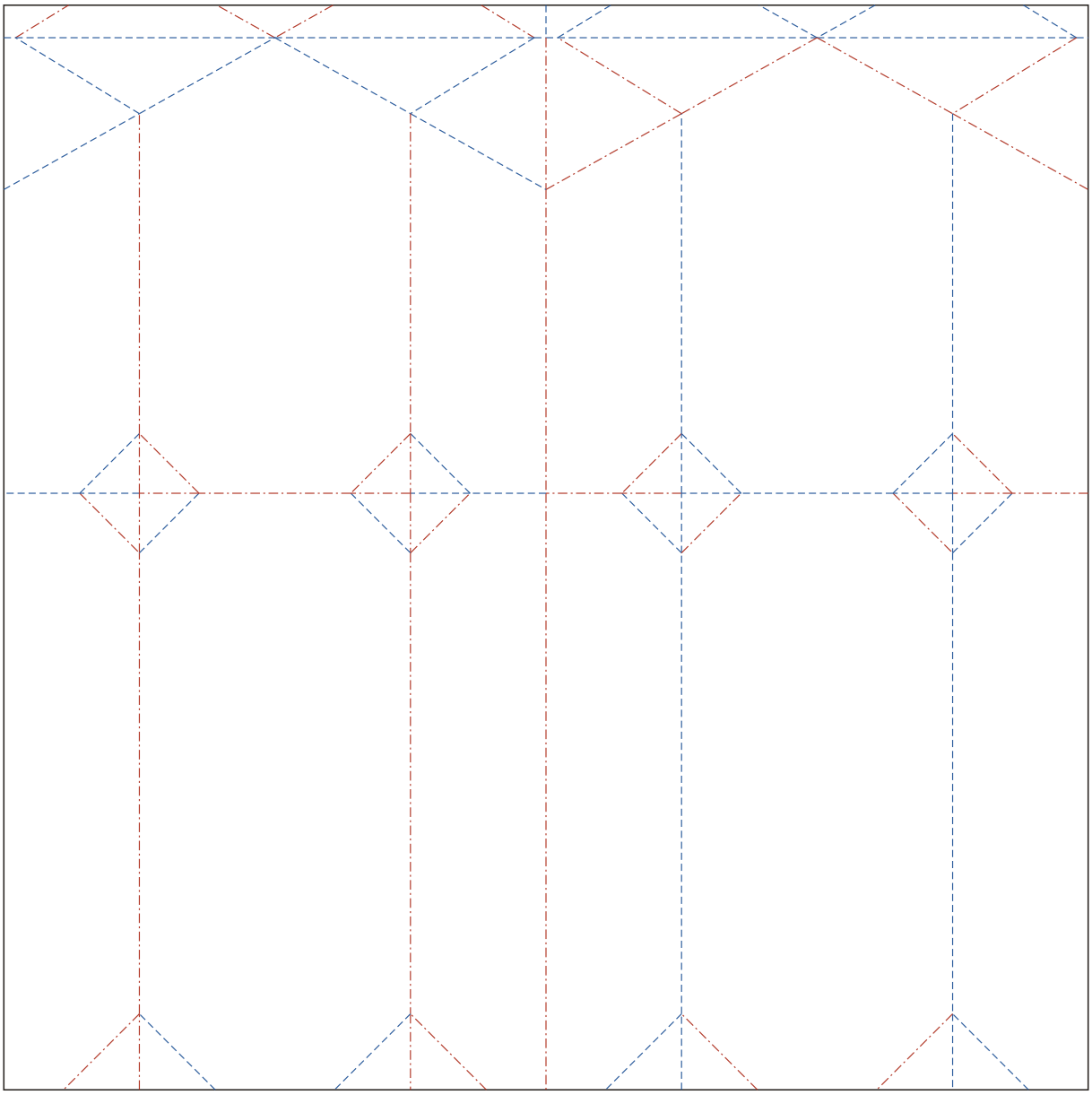
Print side up as drawn. Every crease of the finished piece is drawn as seen from this side, where its layers lie flat.

valley (folds toward you) mountain (folds away)

50 mm: measure this before folding

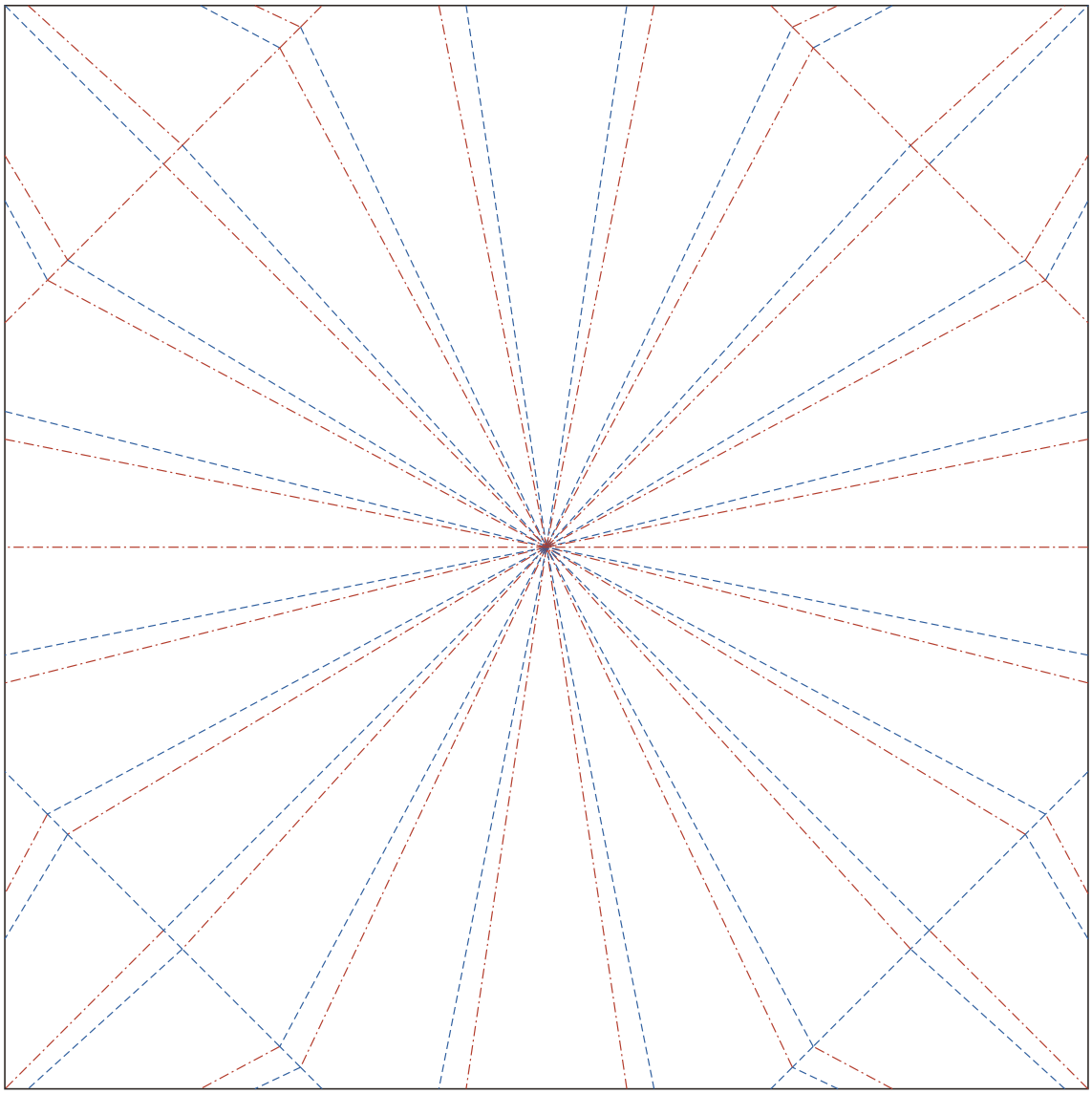
Practice size: print at 100%. Every line is the same pattern at twice the size.

Zori · 16 cm (practice)



Practice size: print at 100%. Every line is the same pattern at twice the size.

Folding fan · 15 cm (practice)



Zori: folding sequence (fold two the same for a pair)

A flat sandal seen from above: a sole with a thong of the paper's other side from a point near the toe out to both sides. Measurements are fractions of the square's side. For a reverse-colour sole, fold the band in step 1 behind instead and turn over before step 2.

1. Print side up. Fold the top edge down by 3.0%: the thong.

2. Fold the right half behind onto the left, edges together. The back half's band is now on its outside.

3. All layers: from the middle of the top edge, fold each top corner down along a line meeting the side 14.0% below the top. The back half's band comes round to the front as two strips from the point.

4. Turn over. Fold both long sides in so the sole is 25.0% wide, centred on the point; the edges meet down the middle.

5. Fold the bottom edge up past the top until, seen from the other side, 13.0% shows above the thong's point: the toe.

6. Fold the four corners in at 45 degrees: 7.0% along each edge at the toe, 5.5% at the heel. Turn over.

Folding fan: folding sequence

An open fan of eleven panels from five pleats a side, all from one rivet point. The pleats are easiest pre-creased on the flat square from the pattern, then collapsed one pair at a time. For a reverse-colour fan, fold step 1 up in front instead of behind.

1. Print side up. Fold the bottom edge up behind to meet the top edge. The middle of the fold is the rivet.

2. Fold each top corner behind along a line 20.7% in from the corner along the top and the same down the side.

3. Pleat from the outside in. On each side, from the rivet: valley-fold along the line 75.93 degrees from the middle and mountain-fold 2.8 degrees further out, so the outer paper slides in over a narrow tuck. Then the same at 59.05, 42.18, 25.31, 8.44 degrees, working inward. Press flat.

What paper has to cope with

The zori: 44 x 20 mm from an 8 cm square, the thong 2.4 mm wide. Up to 16 layers, at the heel corners; 8 over half the sole. Sharpest angle 29 degrees, nothing narrower than 2.4 mm. Thin paper (kami or thinner) helps at the heel.

The fan: 6.62 x 3.88 cm from a 7.5 cm square. Up to 12 layers. The tucks are 2.8 degree wedges meeting at the rivet (about 3.7 mm wide at the edge): pre-crease them; the rivet is where all of them meet, so hold it flat while you collapse each pair. Nothing locks: press the fan under a book.

Both patterns pass the app's flat-foldability and rigid-motion checks. Neither has been folded in paper yet.

If you fold them

Photograph the front and back beside the app's Display view (same paper side up). Note any step that fought you, anywhere the paper buckled or would not lie flat, and whether the thong reached both sides.